High-level Architecture

This lesson gives a brief overview of Cassandra's architecture.

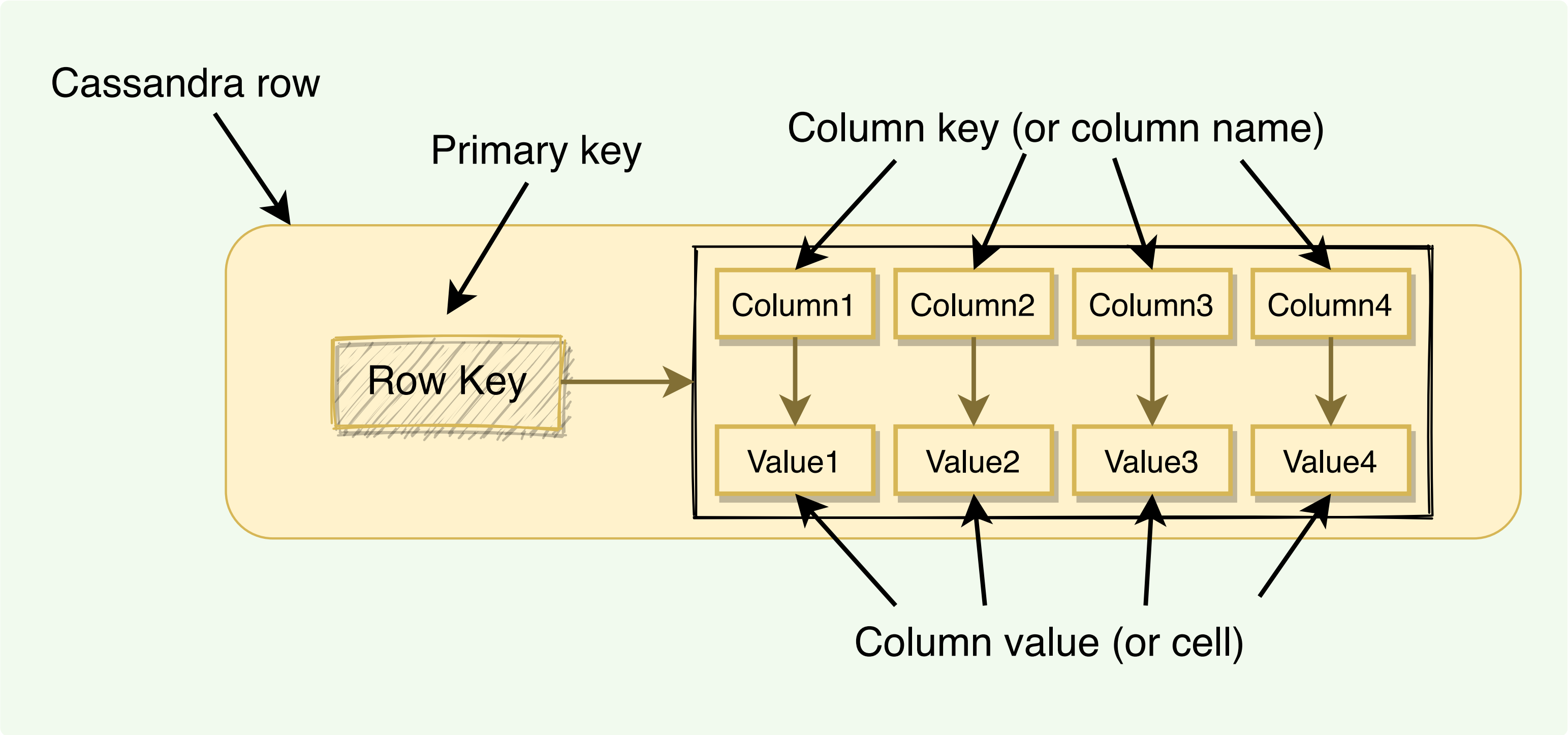
Cassandra common terms

Before digging deep into Cassandra's architecture, let's first go through some of its common terms:

Column: A column is a key-value pair and is the most basic unit of data structure.

- **Column key**: Uniquely identifies a column in a row.
- **Column value**: Stores one value or a collection of values.

Row: A row is a container for columns referenced by primary key. Cassandra does not store a column that has a null value; this saves a lot of space.



Components of a Cassandra row

Table: A table is a container of rows.

Keyspace: Keyspace is a container for tables that span over one or more Cassandra nodes.

Cluster: Container of Keyspaces is called a cluster.

Node: Node refers to a computer system running an instance of Cassandra. A node can be a physical host, a machine instance in the cloud, or even a Docker container.

NoSQL: Cassandra is a NoSQL database which means we cannot have `joins` between tables, there are no `foreign keys`, and while querying, we cannot add any column in the `where` clause other than the primary key. These constraints should be kept in mind before deciding to use Cassandra.

High-level architecture

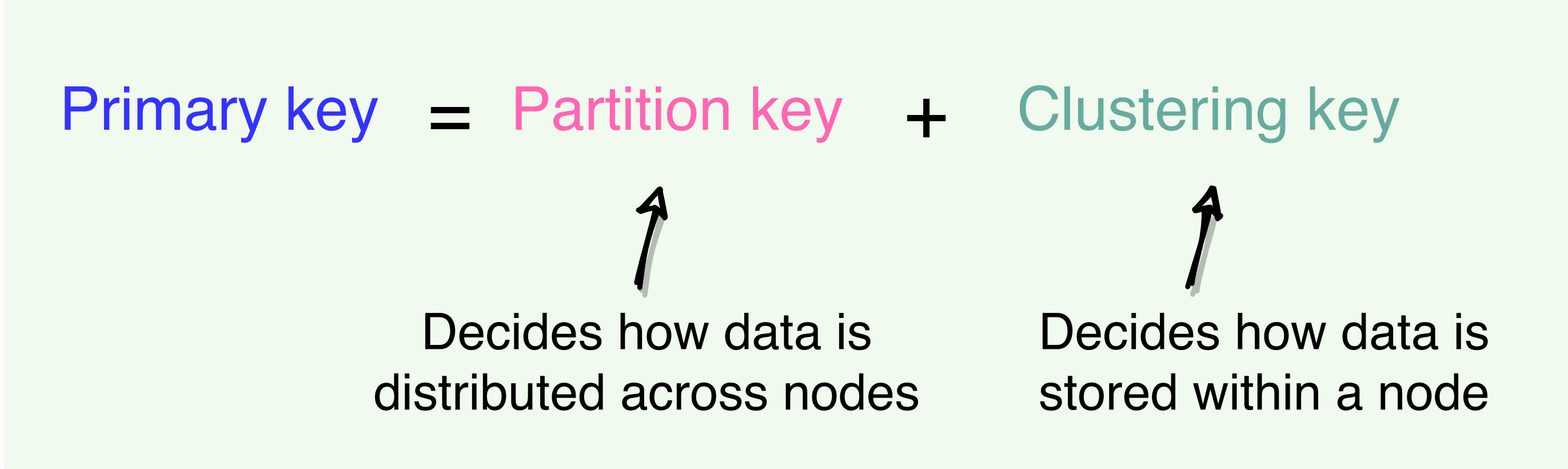
Data partitioning

Cassandra uses **consistent hashing** for data partitioning. Please take a look at Dynamo's data partitioning; all consistent hashing details described in it apply to Cassandra too.

Let's look into mechanisms that Cassandra applies to uniquely identify rows.

Cassandra keys

The **Primary key** uniquely identifies each row of a table. In Cassandra primary key has two parts:


Primary key = Partition key + Clustering key
Decides how data is distributed across nodes
Decides how data is stored within a node

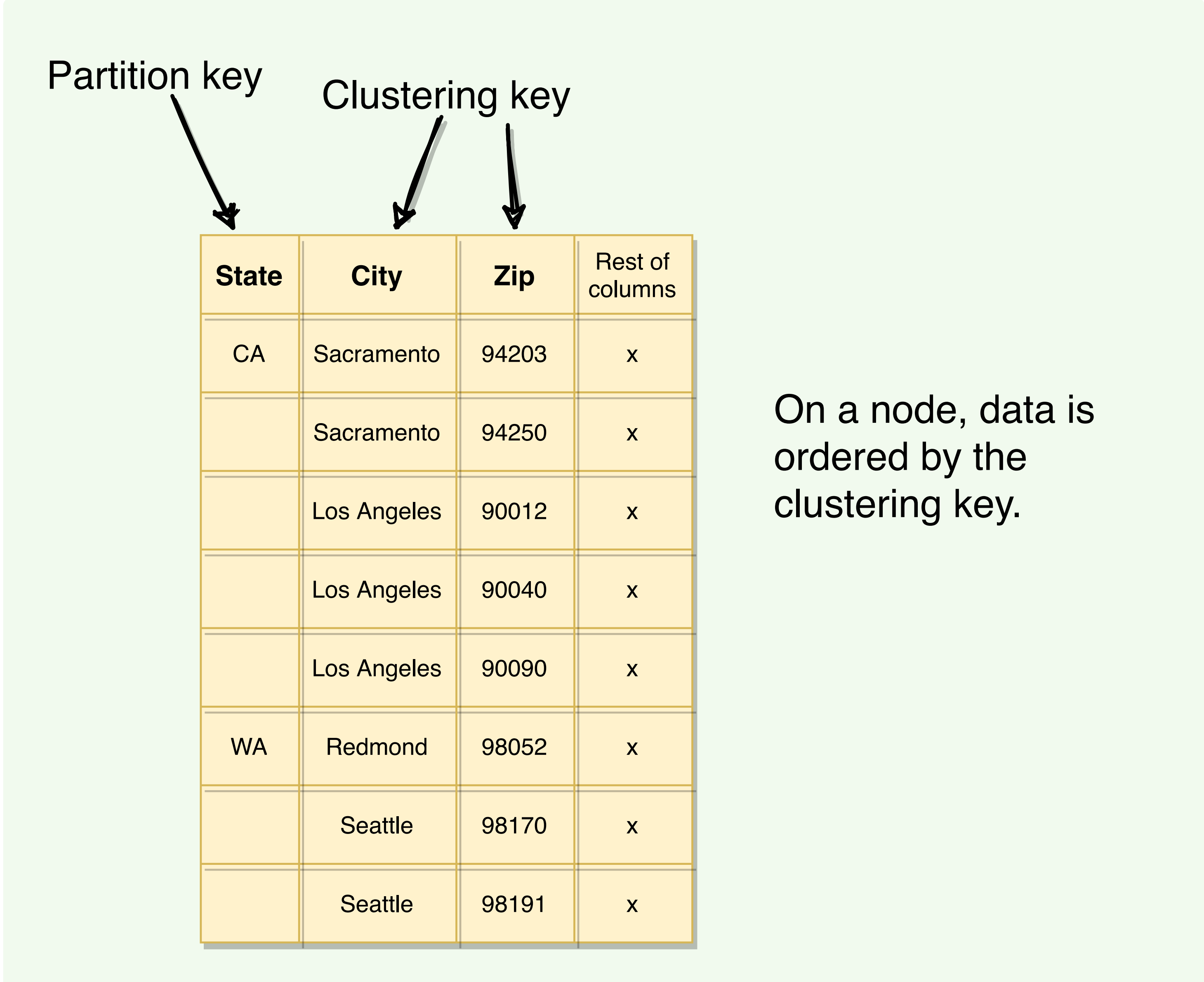
Parts of a primary key

The partition key decides which node stores the data, and the clustering key decides how the data is stored within a node. Let's take the example of a table with PRIMARY KEY (`city_id`, `employee_id`). This primary key has two parts represented by the two columns:

1. `city_id` is the partition key. This means that the data will be partitioned by the `city_id` field, that is, all rows with the same `city_id` will reside on the same node.
2. `employee_id` is the clustering key. This means that within each node, the data is stored in sorted order according to the `employee_id` column.

Clustering keys

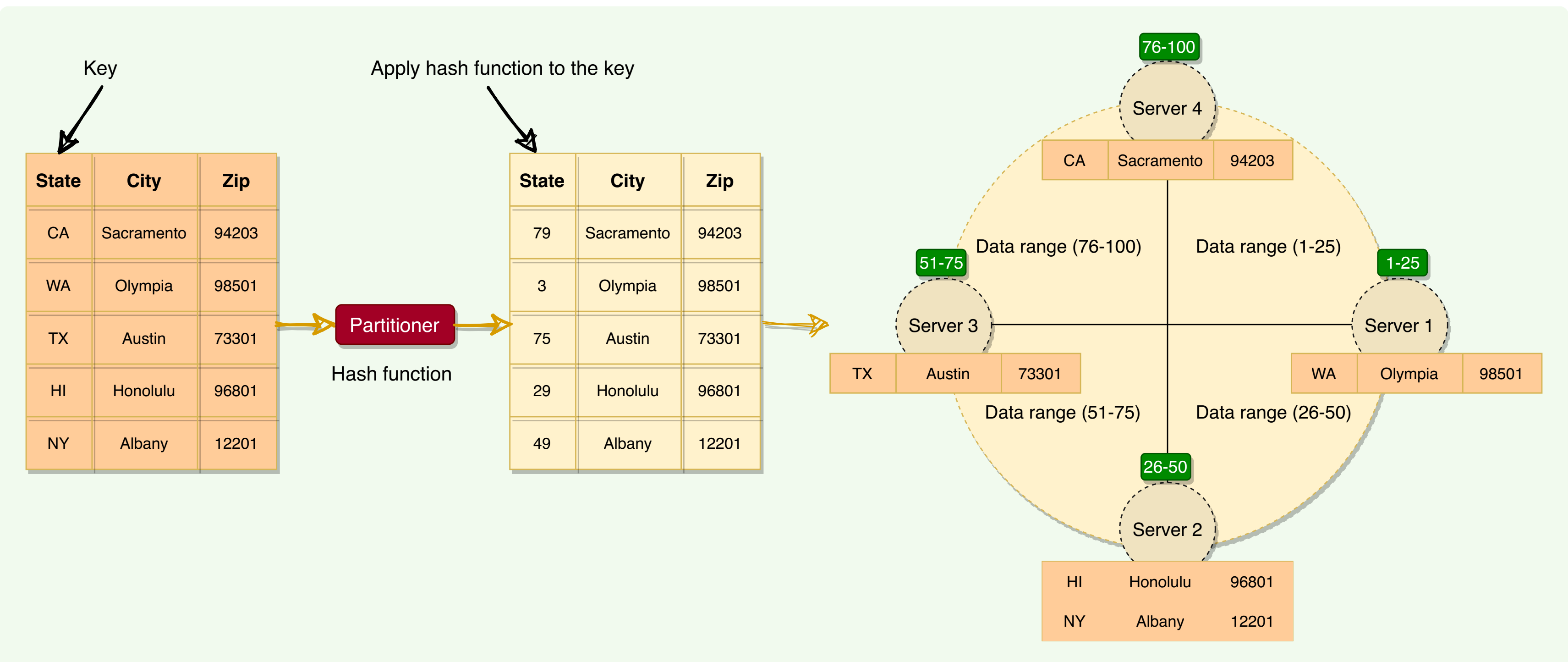
As described above, clustering keys define how the data is stored within a node. We can have multiple clustering keys; all columns listed after the partition key are called clustering columns. Clustering columns specify the order that the data is arranged on a node.



Clustering key

Partitioner

Partitioner is the component responsible for determining how data is distributed on the Consistent Hash ring. When Cassandra inserts some data into a cluster, the partitioner performs the first step, which is to apply a hashing algorithm to the partition key. The output of this hashing algorithm determines within which range the data lies and hence, on which node the data will be stored.



Distributing data on the Consistent Hashing ring

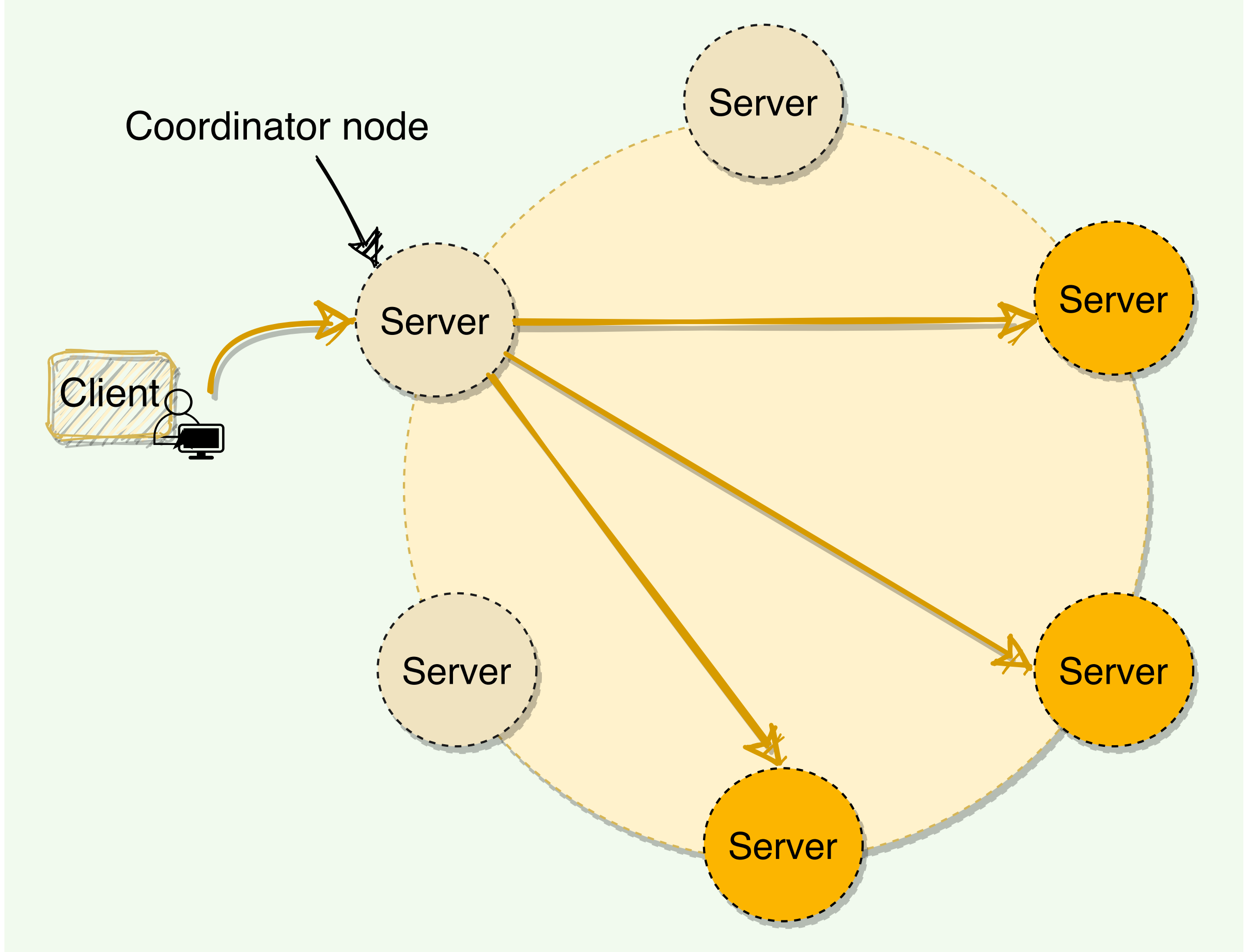
By default, Cassandra uses the Murmur3 hashing function. Murmur3 will always produce the same hash for a given partition key. This means that we can always find the node where a specific row is stored. Cassandra does allow custom hashing functions, however, once a cluster is initialized with a particular partitioner, it cannot be changed later. In Cassandra's default configuration, a token is a 64-bit integer. This gives a possible range for tokens from -2^{63} to $2^{63}-1$.

All Cassandra nodes learn about the token assignments of other nodes through gossip (discussed later). This means any node can handle a request for any other node's range. The node receiving the request is called the **coordinator**, and any node can act in this role. If a key does not belong to the coordinator's range, it forwards the request to the replicas responsible for that range.

Coordinator node

A client may connect to any node in the cluster to initiate a read or write query. This node is known as the coordinator node. The coordinator identifies the nodes responsible for the data that is being written or read and forwards the queries to them.

As of now, we discussed the core concepts of Cassandra. Let's dig deeper into some of its advanced distributed concepts.



Client connecting to the coordinator node

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